Contents

Introduction

1.1 Parkinson's Disease: Background and Burden

Parkinson's Disease (PD) is the most common movement disorder, with 150,000 Canadians estimated to be living with PD by 2034 (Parkinson Canada, 2025; Tysnes & Storstein, 2017). Pathologically, PD is characterized by the loss of dopaminergic (dopamine producing) neurons in the substantia nigra pars compacta (SNc) portion of the brain (Balestrino & Schapira, 2020). This neurodegeneration manifests as both motor and non-motor symptoms. Motor-symptoms encompass tremors, rigidity, instable posture, bradykinesia (slowness of movement), and more. Non-motor symptoms are typically associated with disruptions in sleep, psychiatric conditions, cognitive impairments, gastrointestinal issues, etc. Furthermore, non-motor symptoms cover a broad range of ailments and are seen in approximately 90% of PD patients; however, these symptoms are often harder to recognize than motor symptoms and are thus underestimated when planning treatment strategies (Gökçal et al., 2017).

The clinical diagnosis of PD requires the presence of bradykinesia, assessed using finger tapping or foot tapping movements, and rest tremor or rigidity (Kobylecki, 2020). Studies of clinically diagnosed PD cases demonstrate a diagnostic accuracy of approximately 81%. Common factors relating to PD misdiagnosis are syndromes that manifest similar symptoms such as multiple system atrophy and progressive supranuclear palsy. These conditions, however, display a more aggressive progression and respond poorly to dopaminergic treatment in comparison to PD.

Due to the variable clinical presentation and disease progression among people affected by PD, the disease is commonly categorized into different stages via the Hoehn and Yahr (HY) scale (Carrarini et al., 2019). This scale breaks PD down into five distinct stages, ranging from Stage I (symptoms restricted to one side of the body) to Stage V (wheelchair bound or bedridden) (Carrarini et al., 2019; Clarke et al., 2016). Since its initial publication in 1967, the HY scale has undergone modifications such as the inclusion of half stages to better reflect the complex nature of PD (Goetz et al., 2004). The constant upkeep of this scale has allowed it to remain relevant, easy to apply, practical to both research and patient care settings, and adaptable for both non-experts and PD specialists. However, the HY scale has limitations when it comes to assessing cognitive decline and autonomic nervous system dysfunction common in PD patients. As such, diligent communication between clinicians and patients remains essential in accurately assessing and adjusting therapies for PD management.

1.2 Key Stakeholders in PD Management

Effective symptom management and tracking in PD relies on the involvement of multiple key stakeholders. Patients experience the disease first-hand, and document symptoms and fluctuations that may not always appear during clinical visits. Due to this self-reporting mechanism, patients can be subjected to recall bias that ultimately hinders a clinician's ability to provide optimal treatment (Durmaz Celik et al., 2025). Caregivers are essential pillars of support for individuals with PD; they offer holistic support (Longacre et al., 2025). Caregivers provide meaningful insight toward symptom tracking and it's associated challenges, however this means that caregivers must be constantly alert, creating high levels of burden and potential oversight.

The role of healthcare providers in PD symptom tracking varies depending on their specialization. General practitioners (GPs) describe their role as an extended coordinator of care (Plouvier et al., 2016). More confidence was reported as the disease progressed compared to early-stage PD, specifically when palliative care becomes prominent (Plouvier et al., 2017). Specialists, such as neurologists, possess specific knowledge about PD and oversee PD patient care when self-management is not possible. Issues of accessibility may arise for patients in rural areas who lack access to specialized neurologists (Aslam et al., 2025). Additionally, being referred to a neurologist does not guarantee appropriate PD treatment; general neurologists may not be experienced in movement disorders and thus not recognize the need for advanced therapies (Aslam et al., 2025). Overall, the distinct perspectives and limitations of patients, caregivers, GPs, and specialists demonstrate why PD symptom tracking is a multi-faceted area that requires greater development and cannot be addressed through a single solution.

1.3 Wicked Problem Statement

Tracking symptoms and disease progression is a pivotal aspect of PD management and is typically charted collaboratively by clinicians and PD patients (Heijmans et al., 2019). Most often, this process involves using the Movement Disorder Society Unified Parkinson's Disease Rating Scale (MDS-UPDRS) (Guo et al., 2022; Heijmans et al., 2019). Though this scale provides consistent categories and metrics that clinicians can refer to, there are several limitations: the assessment of progression via the MDS-UPDRS is dependent on visits with clinicians which may be infrequent. Additionally, the scores given by clinicians are subjective and may not reflect the true extent of the disease's progression, patients might inaccurately recount symptoms or demonstrate recall bias (Heijmans et al., 2019). Due to these limitations, a wicked problem emerges in which PD symptoms are inaccurately tracked due to self-reporting biases and infrequent patient visits to clinicians. This leads to impersonalized treatment strategies that can limit the efficacy of symptom management due to the varying ways symptoms manifest across individuals with PD. Through this review, we wish to identify how effectively current PD symptom tracking methods capture individual disease progression.

Biomedical Background of PD

2.1 Epidemiology

Variations in risk of developing PD exist throughout age, sex, race and ethnicity, socioeconomic status (SES), genetics and environmental factors. Age is one of the strongest demographic factors associated with PD, with prevalence increasing among older populations. A world-wide meta-analysis states that the prevalence of PD per 100'000 individuals increased from 0.04% among individuals aged 40-49 years to 1.09% among those aged 70-79 years old (Pringsheim et al., 2014). However, it is still not certain if this association is linear or exponential. Furthermore, aspects of aging that underlie this association such as age-related neuronal vulnerability remain unknown (Ben-Shlomo et al., 2024).

While incidences of PD in women are rising, PD is more common in men by a 4:1 ratio which has been hypothesized to be due to increased exposure to environmental risk factors such as pesticides and occupations with high toxic exposures (Ben-Shlomo et al., 2024). Studies have suggested variation across races and ethnicities which may contribute to development of PD due to penetrance and genetic risk factors associated with PD (Healy et al., 2008). Finally, an estimate of 10-15% of PD cases are due to causative genetic mutations

(Stocchi et al., 2024) including well-established autosomal dominant and recessive mutations which cause PD and parkinsonian pathologies such as REM-behavior sleep disorder, motor fluctuations and dystonia (characterized by involuntary and/or persistent muscle contractions) (Jia et al., 2022).

2.2 Neurobiological Foundations & Pathophysiology of PD

Specific structural declines in the brain are responsible for the cognitive and motor decline seen in PD patients. Key brain structures affected by PD include: the (A) Substantia Nigra, which contains dopaminergic (dopamine-producing) neurons which allow for initiating movement and maintaining muscle tone through synapsing with acetylcholine neurons at neuromuscular junctions (Sonne et al.,2026). Dopamine release from the substantia nigra into the striatum controls initiation and intensity of motor movement, and the scale of atrophy in these neurons is responsible for the severity of motor disability in PD patients (Kordower et al., 2013). (B) Basal Ganglia are a group of interconnected subcortical nuclei and are composed of two main pathways. Within the direct pathway, projections disinhibit the thalamus to facilitate movement, and the indirect pathway suppresses movement (Albin et al., 1989; DeLong, 1990). Dopamine depletion consequently shifts this balance toward net inhibition of thalamocortical drive, producing the bradykinesia and rigidity characteristic of PD. (C) Locus Coeruleus (LC), which is the main norepinephrine output in the brain and loss of LC neurons causes depression, fatigue and resting tremors seen in PD patients (Prakesh et al., 2016). The main cause of loss in these nuclei are the misfolding of the alpha-synuclein protein which is responsible for communication between neurons and misfolding of these proteins leads to their aggregation into clusters known as Lewy bodies which damage neurons and ultimately cause apoptosis (cell death) of neurons which is one of the hallmarks of PD (Spillantini et al., 1997; Hayes, 2019).

2.3 Clinical Presentations of PD

PD presents in individuals as a spectrum of both motor and non-motor symptoms (Bloem et al., 2021). Motor symptoms are easy to identify, and late-stage symptoms include resting tremor, bradykinesia, postural instability, and rigidity (Hayes, 2019). Severe motor symptoms of PD develop due to large atrophy of brain structures, specifically the dopaminergic nuclei of the substantia nigra and these symptoms become noticeable with a delay of around 10 years which build a gap between diagnosis and noticeable symptoms. Earlier symptoms of PD are more difficult to be used as diagnostic factors which include constipations and REM sleep behavior disorder (physically acting out dreams) (Bloem et al., 2021). Recently, non-motor symptoms have become more appreciated and can impact quality of life to the same degree as motor symptoms. Non-motor symptoms include anxiety and depression which are co-morbid in over 35% of PD patients (Reijnders et al., 2008). Interestingly, up to 90% of patients present anosmia (inability to smell) years prior to diagnosis or presentation of other motor or non-motor symptoms (Haehner et al., 2009).

2.4 Diagnosis & Disease Monitorin

Currently many different indicators are used to assess symptoms of PD and are helpful with early identification. These techniques include imaging for cerebrospinal fluid, oxidative stress, inflammation, and neurodegeneration. Studies have focused on identifying biomarkers (an identifying feature that is an indicator of a known biological process) specific for PD as a way to further improve early detection before the onset of symptoms. Prominent biomarkers include glial fibrillary acidic protein (GFAP) which is released by astrocytes in response to damage to the nervous system and has been noted to be a significant indicator of cognitive decline, as well as alpha-synuclein proteins which can be detected and monitored by seed amplification assay (αSyn-SAAs) which is a technique of amplifying and imaging misfolded protein aggregates in the form of Lewy bodies. Other genetic biomarkers have been associated with increased risk for PD and are factors of diagnosing patients with PD. Cohort studies have shown that coupling genetic and blood biomarkers can improve accuracy of diagnosis and can provide better insight into patients' prognosis.

2.5 Current Management & Treatment

There are no current treatments to PD; however, some therapies are effective at alleviating symptoms and improving overall quality of life for patients. Therapies for PD branch into two distinct sections, those aimed at motor symptoms and those for non-motor symptoms. Levo-dopa (L-dopa), a precursor to dopamine, is a widely accepted drug for treating motor symptoms. L-dopa along with other dopamine agonists are highly effective in treating bradykinesia and rigidity and can also help diminish the progression of PD and its associated disabilities. Other drugs such as monoamine oxidase inhibitors (MAOI) have also been used to reduce dopamine degradation within patients and have shown improvement of symptoms (Radhakrishnan., 2018; Stocchi et al., 2024). Treatments for non-motor symptoms include dopamine agonists as well as MAOI but have limited effect as many non-motor symptoms of PD have been hypothesized to have a non-dopamine-related basis (Ahlskog, 2005).

Social and Environmental Determinants of Health in PD

The heterogenous nature of PD manifests as diverse clinical and symptom manifestation, complicating symptom tracking. Beyond just different clinical motor symptoms, one can be affected by social and environmental circumstances. Individual disease progression is influenced by the broader contexts of a patient’s encounters and received care (Van Muster et al., 2024). These determinants of health can affect multiple states of the disease’s course, including its development, how the disease is diagnosed and treated, and ultimately its outcome and impacts on quality of life. Thus, investigating these determinants throughout disease progression provides background to the variance.

3.1 Risks of PD

It is commonly thought that disease development risks are influenced only by physical biological factors: age, sex, and genetics. However, social factors are associated with a higher prevalence of PD (Fan et al., 2026). Notably, variables such as unemployment and food insecurity are correlated with the higher prevalence of PD. These socioeconomic factors may contribute to a population-level burden. However, prevalence variations within populations are often overlooked. For instance, a study showed substantial differences in PD prevalence across multiple provinces in Iran. Here, urbanization rates and annual income per rural household were positively associated with prevalence (Sarmadi et al., 2025). Beyond social circumstances, environmental exposures to chemicals such as pesticides, trichloroethylene (TCE), and air pollution specifically are associated with increased risk of PD. However, this should be accepted cautiously, as disease onset could appear years after exposure (Brolin et al., 2025); correlation, not causation. Geographic and occupational circumstances can exacerbate these risks with increased exposure to these hazards. These findings suggest that social and environmental contexts are important to consider, highlighting the shift from purely biological factors to ones which direct who can develop PD and where in the area it occurs.

3.2 Access to Healthcare, Treatments, and Follow-up Care

Following the disease onset, determinants may create differences in access to a Parkinson's diagnosis, treatment, and follow-up care. For instance, where one lives can influence their access. In Ontario, neurologist visits for PD were higher in areas with greater spatial accessibility, while general family physician visits were more common in remote areas. Moreover, those living in deprived areas had less frequent healthcare visits as their duration of PD increased (Crighton et al., 2024). Yet, restrictions to healthcare are not limited to distance and transportation access. Racial disparities and socioeconomic effects snowball into healthcare inequities. In a study that compared patients of white and black race with PD, those who were black lived in lower-income/lower-education communities, had limited insurance, received healthcare through emergency services, less likely to be on medications for Parkinsonism (Xie et al., 2021). Further than access to services, adherence to non-pharmacological interventions, including rehabilitation, physiotherapy, and speech-language therapy, has been an issue, as personal beliefs, worsening symptoms, and accessibility act as barriers to treatment (Li et al., 2024). Thus, differences in geographic, socioeconomic, and healthcare circumstances may influence not only who receives care, but also the type and extent of care available to individuals with PD.

Relevant Fields of Research

4.1 Understanding Disease Progression

The MDS-UPDRS remains the main way for PD monitoring since it captures motor, non-motor, daily living, and complication domains and has strong overall clinical validity for rating PD (Goetz et al., 2008; Martínez-Martín et al., 2012). However, validity for rating disease is not the same as precision for tracking individual progression.

In early PD, MDS-UPDRS parts II and III are not well targeted, show consistently low effects, and can miss very mild change, which limits sensitivity to progression over short trial windows (Regnault et al., 2019). Real-world longitudinal analyses show that one-year MDS-UPDRS change scores contain substantial error variance, with within-subject reliability ranging only from 0.13 to 0.62; and gait/posture and rest tremor behaving more consistently than many other domains (Evers et al., 2019). Average progression is detectable in cohorts: de novo PD in Parkinsons Precision Medicine Initiative Cohort (PPMI) showed roughly linear worsening over five years, with MDS-UPDRS total increasing by about 4.0 points per year and Part III by 2.4 points per year (Holden et al., 2017). But stage sensitivity differs across measures: UPDRS-II, PDQ-39, and UPSIT tracked progression better than some other common data elements, while cognition and clinician-rated motor scores were less sensitive over 36 months in some subgroups (Lewis et al., 2020).

Non-motor progression is clinically important and heterogeneous. NMSS has psychometric support and can detect longitudinal change across domains, with non-motor burden strongly linked to quality of life (Van Wamelen et al., 2020). Separate non-motor scales can be more sensitive than broad summaries for some changes, but hallucinations, urinary symptoms, cognition, and function repeatedly emerge as strong progression-relevant domains (Bugalho et al., 2021; Dadu et al., 2022)

4.2 Measuring Symptoms Outside of the Clinic

Wearables and smartphone-based tools address the main weakness of clinic scales as they measure symptoms repeatedly in real-world settings rather than sparse clinic snapshots (Daalen et al., 2024; Oyama et al., 2023; Powers et al., 2021; Reithe et al., 2025).

In early-stage PD, remote smartwatch examinations showed moderate-to-strong correlations with consensus MDS-UPDRS ratings for rest tremor, bradykinesia, and gait, good-to-excellent monthly test-retest reliability, sensitivity to dopaminergic medication, and low dropout over one year (Burq et al., 2022). Large app- and smartwatch-based studies also show excellent adherence and strong short-term reliability. The Roche mobile platform achieved 96% adherence and median ICC around 0.9 for prespecified sensor features, with correlations to corresponding MDS-UPDRS items (Lipsmeier et al., 2022).

Several longitudinal studies suggest digital measures can be more sensitive than corresponding clinical items for early progression. In WATCH-PD, gait declined and tremor time increased over 12 months, and some digital changes were larger than changes in corresponding MDS-UPDRS items (Adams et al., 2024). A focused bradykinesia analysis from the same program found digital composites outperformed corresponding MDS-UPDRS items for one-year change detection (Czech et al., 2024). In a multicentre real-world mobility study, digital features had only moderate correlations with MDS-UPDRS, yet showed larger effect sizes for change over time than any MDS-UPDRS part, implying they capture partly different and progression-relevant information (Mirelman et al., 2023).

Gait and fluctuation domains are the most mature. Wearable gait systems objectively track spatiotemporal abnormalities and longitudinal change (Schlachetzki et al., 2017). Tremor algorithms now show cross-device generalizability with high specificity and excellent weekly reliability (Timmermans et al., 2025). Dyskinesia and motor fluctuation platforms can match expected treatment responses and sometimes reveal management opportunities missed clinically (Powers et al., 2021; Antonini et al., 2023).

Non-motor remote tracking lags motor tracking. Wearables show promise for sleep, autonomic function, constipation, and REM sleep behavior disorder, but most non-motor digital biomarkers remain insufficiently validated in Parkinson's-specific longitudinal cohorts (Daalen et al., 2024; Van Wamelen et al., 2021).

4.3 Predicting Individual Disease Trajectories

Biological markers of underlying neurodegeneration remain less mature than clinical and digital symptom measures for tracking individual progression. Conventional CSF total α-synuclein changes early in disease, but longitudinal studies found it did not correlate with motor progression or dopaminergic imaging change (Mollenhauer et al., 2019). Reviews repeatedly conclude that alpha-synuclein is promising but not yet robust enough as a standalone progression marker because of assay, specimen, and standardization problems (Ganguly et al., 2021; Magalhães & Lashuel, 2022; Atik et al., 2016). This is reinforced by a recent PPMI analysis in which baseline α-synuclein SAA parameters did not consistently improve 5-year prediction of motor or cognitive outcomes (Belete et al., 2025).

Machine-learning models for individualized trajectories are advancing faster than wet biomarkers. Several cohorts identify reproducible slow, moderate, and fast progressor groups and achieve good-to-high predictive accuracy from multimodal longitudinal data (Dadu et al., 2022; Salmanpour et al., 2021). Models that incorporate motor and non-motor clinical data, imaging, and sometimes serum neurofilament light outperform single-modality approaches (Dadu et al., 2022; McNamara et al., 2025; Leung et al., 2021; Makarious et al., 2021). Many are constrained by treatment confounding, limited long-term follow-up, unequal training sets, small external datasets, or lack of real-world deployment (Lian et al., 2024; Dehghanghanatkaman, 2025).

4.4 Translating Research into Personalized Care

The field is moving from broad symptom scoring toward multimodal, individualized progression profiling. Precision-medicine frameworks argue that heterogeneous trajectories and treatment responses require subgrouping or patient-level modelling rather than one-size-fits-all tracking (Espay et al., 2017; Marras et al., 2024). In practice, the most realistic model for better capture of symptoms is a combination of scales and digital innovations. Clinical scales remain necessary for interpretability and regulatory familiarity; digital measures add temporal resolution and ecological validity; biomarkers and multimodal machine learning improve prognostic stratification where validated (Rábano-Suárez et al., 2024; Fröhlich et al., 2022; Van Halteren et al., 2020). Two concrete shifts stand out. First, progression assessment is moving from total scores toward domain-specific or weighted composites, including gait-focused measures, milestone outcomes, and stage-specific composites that outperform conventional totals for sensitivity (Brumm et al., 2023; L'Italien et al., 2025; Lo et al., 2022; Regnault et al., 2025). Additionally, remote sensing is making individual fluctuation visible between clinic visits, which aides to personalized care management and proactive monitoring (Burq et al., 2022; Crowe et al., 2024; Van Halteren et al., 2020).

Summary

Parkinson's disease is a multi-faceted disease which is highly heterogeneous and involves individualized clinical manifestations of symptoms. Throughout this review, current methods for symptom tracking in PD demonstrate the ability to capture acute symptoms but fail to follow through with longitudinal progression of motor and non-motor symptoms. Symptom tracking with digital tools such as wearable devices and smartphone tracking can aide in improving PD symptom detection sensitivity and longitudinal tracking as it provides continuous tracking without burden on patients. Recent innovations to PD personalization and symptom tracking include looking at domain-specific scores between PD symptoms rather than total scores evaluated by MDS-UPDRS and incorporating multi-modal biomarkers instead of single modality biomarkers. Going forward, a systematic investigation of the emerging and current methods of symptom tracking in PD will be conducted to identify the most comprehensive PD tracking methods.

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